

Lab 02 - Siemens S7 PLC Emulation



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Lattes

ORCID

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Scopus

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substationworm

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Tasks

Warning:

The tasks in this document are strictly educational, observational, and non-intrusive, and must fully comply with ethical and legal standards.

- **1** Verify the IP address of the `otlab-student` workstation.
- **2** Determine the subnet range of the network where the `otlab-student` workstation is deployed.
- **3** Discover the IP address, MAC address, and vendor information of other active hosts within the network.
 - *Hint: It is a PLC.*
- **4** Identify open ports and available services on the OT-ICS host over both TCP and UDP protocols.
- **5** Determine the proprietary industrial communication protocol used by the PLC.
- **6** Retrieve additional system information using vendor-specific `nmap` scripts.
- **7** Execute a `plcscan` on the OT-ICS device detected on the network and collect further data.

Note: On the `otlab-student` workstation, the `plcscan` tool ([meeas/plcscan](#)) must be executed using `python2`.

Tools

- The following tools are available on the `otlab-student` workstation for completing OT-Lab 2: `ifconfig`, `masscan`, `netdiscover`, `nmap`, and `plcscan`.

Nomenclature

- ICS: Industrial control system.
- IP: Internet protocol.
- MAC: Media access control.
- OT: Operational technology.
- PLC: Programmable logic controller.
- TCP: Transmission control protocol.
- UDP: User datagram protocol.